JIRAMET WATTANAPHAIBOON

AI Engineer | Sensor Data & IoT Systems

AI / IoT Engineer with hands-on experience building applied AI systems from data exploration and model development to backend integration and evaluation. Developed an E-Nose analytics system for odor classification using multi-sensor time-series data, and enjoy translating sensor data into practical AI applications for real-world deployment.

(+66) 81-123-0463 | Suan Luang, Bangkok 10250, Thailand | sendtocartoon@gmail.com

EDUCATION

KASETSART UNIVERSITY
B.Eng. Computer Engineering
GPA: 3.48/4.00

SOFT SKILLS

- Problem-solving
- Analytical thinking
- cross-functional communication
- self-directed learning

LANGUAGES

Thai (Native)

English (Working proficiency)

ADDITIONAL PROJECTS

- Speech Emotion Recognition (Thai SER) — Deep Learning
- Multi-label Movie Genre Classification (TMDB 5000) — NLP
- Movie Rating/Success Prediction (MovieLens) — XGBoost

WORK EXPERIENCE

DRONE ACADEMIC (THAILAND) CO., LTD. APR 2025 – JUNE 2025

AI Engineer Intern

- Benchmarked RTX 4090 (24GB vs 48GB) with Qwen 2.5 (7B/14B/30B, FP16), simulating concurrent users (1–16) and tracking OOM, latency, and throughput.
- Built the backend for an AI Avatar pipeline: integrated Whisper for speech-to-text (STT), routed transcripts into an LLM for response generation, and synthesized audio replies using F5-TTS (TTS), delivering an end-to-end conversational voice loop.
- Fine-tuned a Thai TTS voice model (F5-TTS) by extending the original training setup with additional Thai voice data and targeted samples, improving pronunciation clarity for specific Thai words that were previously unclear

PROJECTS

AI MATH — INTELLIGENT TUTORING SYSTEM (LLM/RAG/VISION)

Developed an AI tutoring platform that supports textbook-grounded Q&A, worksheet generation, and automated grading from PDF/image submissions with Vision/OCR and partial-credit feedback; built with pgvector-based RAG + reranking, FastAPI, and Redis worker queues

E-NOSE ANALYTICS — ELECTRONIC NOSE ODOR CLASSIFICATION (PYTORCH / STREAMLIT / TIME-SERIES SENSORS)

Developed an E-Nose analytics platform for odor classification using multi-sensor time-series data; built interactive visualization and inference workflows with Streamlit, and trained/evaluated CNN, LSTM, and Transformer models in PyTorch to compare classification performance across sensor patterns

SKILLS

- LLM/GenAI: RAG, Embeddings, Reranking, Prompt Engineering, LoRA/PEFT, LLM Evaluation
- ML/Data Science: Data preprocessing, EDA, Feature engineering, Model evaluation; scikit-learn, TensorFlow
- Speech AI: Whisper (STT), F5-TTS (Thai voice fine-tuning), TTS pipeline
- Vision/Docs: OCR, PDF/Image ingestion, handwriting understanding (workflow)
- Backend: Python, FastAPI, REST APIs, Async workers/queues
- Data: PostgreSQL, Supabase, pgvector, Redis
- Infra/Tools: Docker, Docker Compose, Git/GitHub
- Productivity: Excel, Google Sheets

JIRAMET WATTANAPHAIBOON

AI Engineer

AI Engineer with hands-on experience building and deploying AI solutions end-to-end—from requirement exploration and PoC to backend integration and system evaluation. Strong in LLM/RAG + Vision pipelines and performance benchmarking. Enjoy translating business needs into practical AI features and continuously improving models in production.

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EDUCATION

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SOFT SKILLS

- Problem-solving
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- cross-functional communication
- self-directed learning

LANGUAGES

Thai (Native)

English (Working proficiency)

CURRENT PROJECTS (IN PROGRESS)

- ComfyUI character-consistency workflow for future video generation.
- Reinforcement Learning study via a custom 3D Connect-4-with-gravity game environment.

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ADDITIONAL PROJECTS

- Speech Emotion Recognition (Thai SER) — Deep Learning
- Imbalanced Tabular Classification with ADASYN/SMOTE
- Multi-label Movie Genre Classification (TMDB 5000) — NLP
- Movie Rating/Success Prediction (MovieLens) — XGBoost
- Fashion Item Classification (Fashion-MNIST) — MLP vs CNN

SKILLS

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